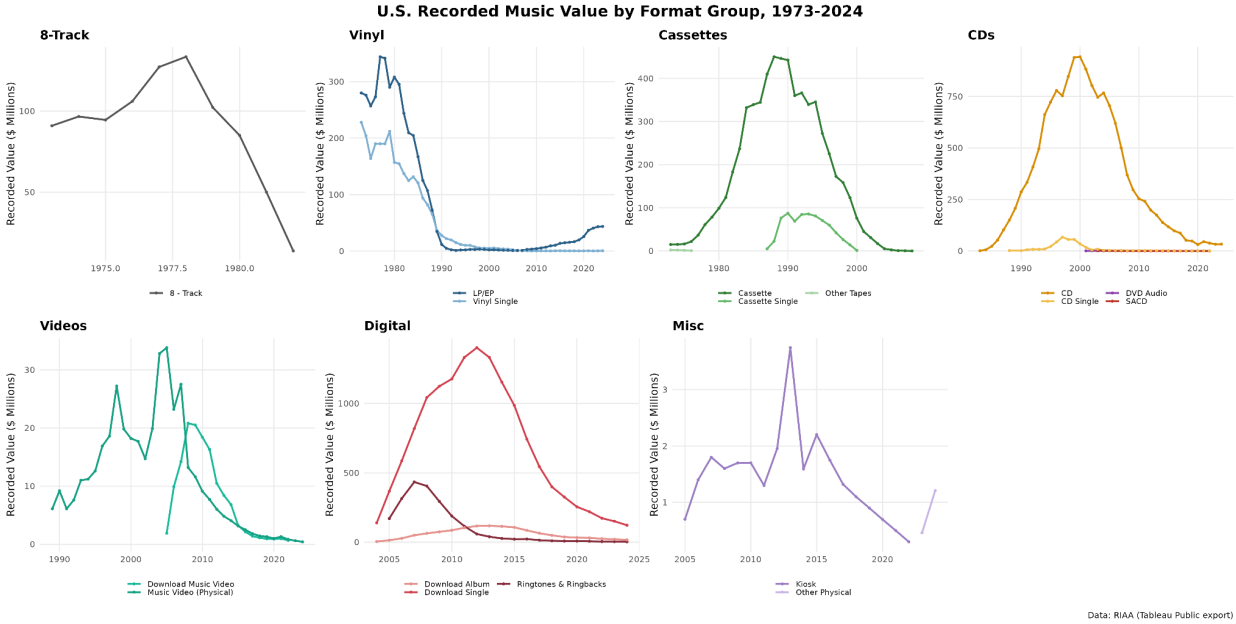
With streaming seemingly being today's go-to way of consuming music, one has to ask, are the alternative methods a relic of the past and on their way out? And for any traditional methods that remain, are they also on their way out? What does the lifespan look like in general, after its peak? We investigate these questions, and also look into if these formats all follow a similar predictable lifecycle.

This analysis uses the U.S. recorded music sales data available through an RIAA Tableau Public export, covering 1973 to 2024 and 17 recorded formats. The analysis examines how formats change over time by comparing their historical peaks, subsequent declines, and recent trajectories. To measure severe decline, we define a benchmark as the first year in which a format's recorded value falls to 10% of its historical peak. We also compare formats using growth rates, full-history trend patterns, and correlations in annual changes. Because not all formats span from 1973 to 2024, the results are interpreted as comparisons of recorded-value trajectories rather than definitive measures of format survival or consumer demand. The analysis covers the categories present in the export and should not be treated as a complete measure of the entire recorded music market.

We first start off with the raw data. For clarity's sake, we group the formats into 7 categories, with Misc representing the only 2 formats without a clear grouping:



The speed of declines:

For every format, we found the peak year/value and then counted how many years it took to fall to 10% of that peak. We defined the -90% threshold as an objective, consistent yardstick that includes as many formats (since 4 formats haven't met this threshold, whether due to coverage stopping before reaching such lows, or being too recent), so formats could be compared on equal footing. On its own, this uncovered that Download Single collapsed to -90% of its peak in 12 years, faster than Cassette (13) and much faster than CD (17).

To be precise, the average years needed for physical formats was 10.3, whereas digital was 9. This could lead into the theory that digital platforms did not buy durability, and that physical platforms, perhaps due to sunk-cost sentiment since digital platforms didn't require any friction.

The lifespan of formats:

We'd also like to ask if every format follows a similar predictable lifespan, one might naively think a bell curve or "textbook" (a curve taking similar times to rise and fall) would be the default. Looking at the graphs, there are examples that support this, like CD (17 yrs up, 24 down) and Cassette (15 up, 20 down).

There are however other curves, particularly those that spike sharply or start out at / near their peak, then fade, with Ringtones/Ringbacks (2 years to peak, then 17-year tail) and DVD Audio (4 up, 17 down), Download Music Video (3 up, 14 down), CD Single (9 up, 25 down).

Vinyl Single and SACD start at their peak.

U-shaped / partial revival examples break the mold entirely: with LP/EP peaking in 1977 at $344M, then cratering for ~30 years, and has been climbing back since ~2007. Cassette single also went through a relatively similar experience, with it quickly coming near its peak of 1990 after 3 years.

But instead of eyeballing which formats share a shape, we also let an algorithm find the groupings on its own, using measurements like how fast each format climbed to its peak, how long it took to fall afterward, how large it got, and when in history it happened.

The cleanest result came from asking the simplest version of that question: with no number of groups suggested in advance, what's the single most natural way to split all 17 formats? The answer, consistently, was just two groups: Vinyl Single and LP/EP in one, all other 15 formats in the other. That's not a soft "vinyl looks a little different." Of every possible way to divide this entire market, the most defensible one is simply vinyl against everything else.

Asking for more detail than that breaks the "everything else" side into pieces that make intuitive sense: a cluster of small, always-niche formats that spiked and vanished fast (Kiosk, DVD Audio, SACD), the earliest-era formats from the 1970s (8-Track, Other Tapes), and one large cluster covering every format that was ever a genuine mainstream purchase mechanism, whatever decade it happened in.

We also ran the same exercise using only how long each format's rise and fall took: no dollar amounts, no sense of what year it was. That version draws a sharper line between the "long, slow legacy" formats like CD and Cassette and everything shorter-lived, but the vinyl pairing holds up there too.

Two further checks, using different methods and different rules for picking the number of groups, backed the same conclusion. One, given no target number of groups at all, chose three on its own and still kept the vinyl pair together, apart from everything else. Another built specifically to resist one unusual format throwing off the result went even further: it found Vinyl Single and LP/EP each so unlike anything else that neither one matched even the other, let alone the rest of the market.

So this isn't just "vinyl looks eccentric on a chart." Four separate methods, built on different assumptions and given different amounts of guidance, all landed in the same place: vinyl hasn't moved with the rest of the market. And as we'll uncover later, it will likely continue its own path.

How formats interact and correlate:

Of course, none of these formats exist in a vacuum, the time and conditions allow formats to thrive or could stunt their growth, we try finding the relations between formats.

There are 17 formats, meaning 136 possible pairings, some of these don’t exist since some lifespans simply do not match up even for a year. Accounting for that, we end up with only 82 pairings then. That filter itself would still be too generous, since it doesn't check whether both formats had real dollar values during the specific years they overlapped (e.g., Vinyl Single's peak was 1973, so by the time it overlaps with Ringtones/Ringbacks in 2005-2024, Vinyl Single is already severely down, passing the filter on a historical technicality while still being noise in practice).

Applying that stricter, more honest lens by hand to the list: The results are mixed, and the pattern itself is a finding, running year-over-year correlations between formats' dollar changes:

Download Single and CD moved in opposite directions for their entire two-decade overlap, from 2004 to 2024, and both were multi-hundred-million-dollar formats the whole time. That's the one real, unambiguous substitution in the whole dataset — one grew because the other shrank.

Download Album and CD show the same handoff, just smaller money on both sides.

Download Single and Download Album is the strongest relationship anywhere in this data, and it runs the other way entirely: they rise and fall together. Not competing for the same dollar, just two symptoms of the same thing, people buying digital.

Ringtones and Download Music Video pair up the same way, smaller stakes, same mobile-era money moving both places at once.

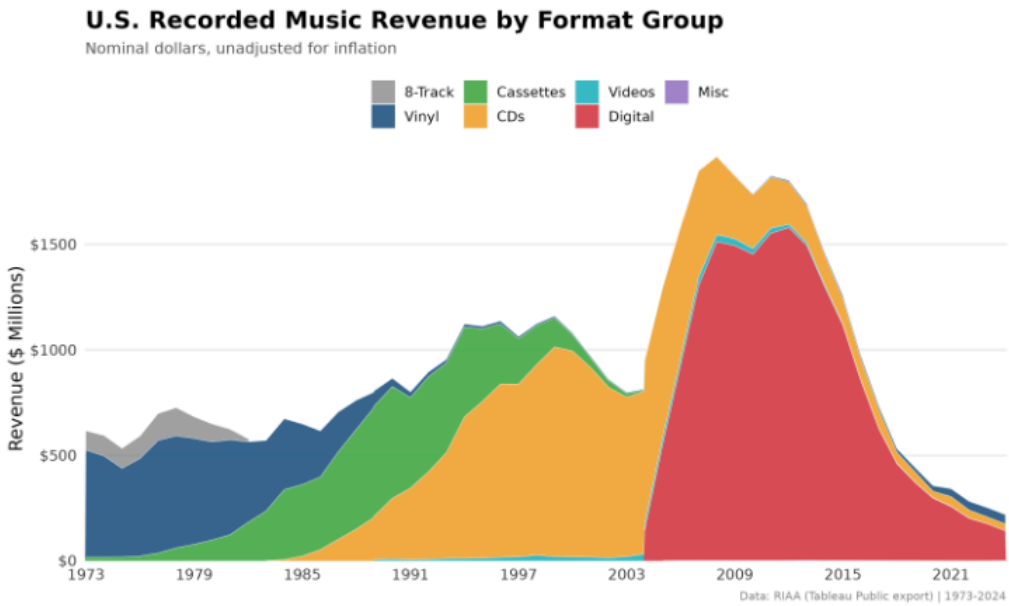
Then there's the stuff that looks dramatic and isn't. Cassette Single against Vinyl Single, SACD against Cassette, Download Music Video against SACD — every one of these involves a format already down to pocket change during the years being compared. A format doing $100K one year and $300K the next can throw off a correlation that looks alarming on a page and means nothing in a bank account.

Circling back to the initial question: CD and Download substitution, Download Single and Album complementarity are in fact the only economically meaningful relationships in the data, and everything else that looks dramatic is a small-numbers artifact.

Also worth knowing: in raw dollars, unadjusted for inflation, the total pie (all formats summed) actually grew from 616M (1973) to a peak of 1.9B (2008) before collapsing to $220M by 2024. So for most of this history, format transitions weren't zero-sum reshuffling: the total market expanded even as one format replaced another. The pie only started shrinking in the last ~15 years, which is really money leaving this whole category for other ventures not captured in this dataset, not formats cannibalizing each other. This complicates the naive narrative as digital formats kill physical ones further.

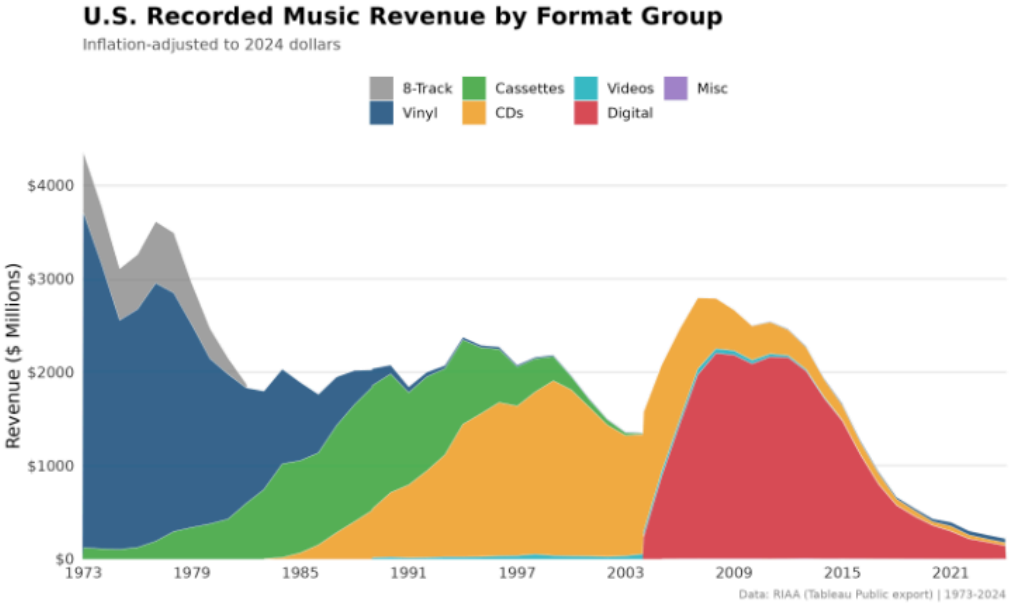
Inflation:

Expanding on the previous section, looking at the unfiltered data, the story looks like growth. Total revenue across these formats climbed from $616 million in 1973 to a peak of $1.9 billion in 2008, and it seems like each new format simply added to the pile rather than taking from the one before it. That's the chart below, using the similar groupings from the earlier visualizations.



The next chart tells a different story. A dollar in 1973 bought far more than, say, a dollar in 2008, so adjusting every year into 2024 dollars changes the shape completely. Measured that way, the market's tallest point isn't 2008. It's 1973, the very first year in the data, at roughly $4.4 billion. The 2008 peak, which looked like a record in raw dollars, comes out to about $2.8 billion, still well short of where the market started. From that 1973 starting point to 2024's $220 million, the real decline is about 95%. Even from the 2008 high, it's about 92%.

So the format transitions weren't expanding the market after all. They were slowing a long slide, and the digital boom of the mid-2000s was a partial recovery, not a new high. The same goes for individual formats: CD's 2000 peak of $942 million is worth about $1.7 billion in today's money, a figure the raw chart badly understates.



But can we go further? What can we say about the future of the remaining formats?

Being able to cluster these formats is good and all, but can we use this to see how the remaining formats behave in the future?

One could assume no new competing formats get introduced, although this is dubious, given formats are still being introduced as late as 2023 with "Other Physical".

Grouping the 8 formats still standing in 2024 by how they're actually behaving shows four distinct patterns, not one uniform slide to zero:

The clean deaths: Ringtones, Download Album, Music Video (Physical) decay so consistently that "heading to zero" isn't speculation, it's just arithmetic on an already-tight line.

CD is dying, but messily: It's bumpier, echoing what we found earlier: CD's fall has always been slower and less orderly than the digital formats that came after it. It'll likely still exist in some small form for a while.

Vinyl Single has basically stopped dying: its confidence interval crosses zero, meaning it is genuinely provably stable, not on a constant decline. It found a tiny, stable floor of hardcore collectors buying specific 7" singles.

Download Single, while still falling around 20%/year, stays the largest of the bunch in dollar terms even five years out. The one format breaking every part of this pattern is LP/EP.

Speaking of, LP/EP is growing: this format breaks "they're all destined to disappear" as a thesis. It's not a survivor limping along, it's the one genuinely expanding line in the entire "still alive in 2024" list. The caveat with this being that it was fit with a plain exponential curve, which assumes unbounded compounding growth forever. Real markets don't work that way — vinyl production is famously capacity-constrained by a small number of pressing plants worldwide, so naive extrapolation to $94M by 2029 is probably too aggressive. When we instead let growth level off toward a ceiling, the data fit that shape far better and put 2029 closer to $45–60M. LP/EP's growth has already slowed from 44% in 2021 to just 1% in 2024.

Outliers, what do they have to say?

Circling back to the idea of formats not existing in a vacuum, we should investigate whether there are any outliers worth flagging, after all, in the previous section, our CD fit had its decline look noisier than the digital formats, and 2020 is sitting right in that window — a year physical retail was disrupted by COVID.

Here is what we find: For most, the answer is no. CD is the clearest example of why that check matters: 2020, the year COVID shut down physical retail, does show up as a real anomaly in CD's history — but removing it barely moves CD's long-run decline rate at all. It means CD's messier, bumpier fall compared to the digital formats around it is a real, structural difference, not an illusion created by one pandemic year.

Vinyl Single turns out to be the format most worth re-examining — but not in the direction you'd expect. Looking at its actual full recovery (all the way back to when it stopped shrinking after 2008) tells a more solid story: no single year is propping this up. Vinyl Single really has found something close to a floor, and it's held for sixteen years running, not simply a lucky break.

One other format is worth flagging for the opposite reason. Cassette Single's reported collapse rate turns out to be the least trustworthy number in the entire dataset — a single year at the very end of its run swings its estimated decline by more than ten percentage points. It's a reminder that a short-lived, niche format's "trend" can be far more fragile than it looks on the surface, even in a dataset that is thoroughly checked.

Limitations:

There was a general trend throughout analyzing this dataset that the amount to work with was often too small for some aspects, for instance, one could conclude digital purchase formats decline faster in dollar terms once the decline starts, but we cannot reasonably conclude they necessarily also disappear just as fast, or even faster, than physical.

We also tested standard product-lifecycle curves against the simpler trend lines used above. They agreed on every direction, but not on how much money is left: Download Single's 2029 figure ranges from about $11M to $47M depending on the model, and the data can't pin LP/EP's ceiling tighter than roughly $45–90M.

For the inflation section, one must be cautious: because 1973 is where this dataset begins, we cannot say the market was never larger before then, only that it was largest at the start of what we can see. And the earlier finding about how fast formats collapse holds up either way: in inflation-adjusted terms, Download Single still fell to a tenth of its peak in 10 years, ahead of Cassette (13) and CD (16).

Conclusion:

So, are these formats just relics on their way out? Well some are, some aren't, and the ones that survived didn't survive for the same reason.

Three of the eight formats still standing in 2024 are dying on a schedule clean enough to set a watch by. CD is dying too, just messily, the way it's always fallen — slower and less orderly than anything digital. But Vinyl Single and LP/EP break the pattern outright. One found a small, stubborn floor and stopped falling. The other is the only format in this entire dataset still genuinely growing, sixteen years into its own comeback. Every method we threw at this data kept landing on the same conclusion: vinyl doesn't behave like the rest of the market, and it hasn't for a while.

That alone should complicate the easy version of this story, the one where digital arrives and physical just dies. It's messier than that. Digital collapses faster once it starts falling — that part held up everywhere we looked — but it doesn't buy that speed for free; it just trades the slow bleed of a format people already own for a much sharper drop the moment something better shows up. And the market itself wasn't quietly growing the whole time, the way the raw dollars suggest. Adjust for what a dollar was actually worth, and the real high point is 1973, not 2008.

None of this is about streaming, and that's on purpose. Streaming isn't in this dataset, and it didn't need to be for the story to hold. The decline started, and the market changed shape, well before streaming ever entered the picture — which says something on its own about how much weight that one explanation is usually given.